



An estimation method of smart meter errors based on DREM and DRLS

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ABSTRACT

Remote estimation of smart meter errors based on measurement data analysis from the user smart meter comes into focus because field calibration has high maintenance cost and difficult to analyze in time. In this paper, a remote estimation method of smart meter errors based on Dimension Reduction Estimation Model (DREM) and Damped Recursion Least Squares (DRLS) is proposed, in which DREM deals with the insolvability of actual model, and DRLS algorithm improves the stability and accuracy of estimation. To verify the effectiveness and practicality the method proposed is applied in both laboratory and actual distribution feeder unit. The results show that the proposed method did not need to calculate the network loss independently in advance, and it can estimate the smart meter errors and network loss rate in real-time. Finally, the important parameters involved in the model and algorithm are discussed, and the advice of parameter selection is proposed.

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1. Introduction

Smart electricity meters are an important part of smart power grid, and they are also the basis for power grid operation control and trade settlement between power supply and electricity consumption. The measurement results are directly related to the security of power grid and whether the trade settlement between both sides is fair, fair and reasonable. Therefore, the operation status evaluation of power metering devices is particularly important. The operation reliability of smart meters not only affects the income of power grid companies, but also has a direct bearing on the interests of millions of electricity users [1]. At present, practical theory and technology that could accurately measure and monitor the operational smart meter errors in real-time are immature in power industry. China has installed more than 500 million smart meters [2], and traditional calibration to such a large quantity of smart meters would be costly, which is realized by installing millions of on-line monitoring equipment or dispatching maintainers frequently. To reduce the measurement cost for power companies and society, the remote estimation of smart meter errors based on the analysis of large-scale measurement data becomes a research

focus [3].

Some achievements have been obtained in the calibration study of smart meter errors. The calibration method of smart meter dynamic errors is detected by the test system based on pseudo-random distortion test signal and indirect likelihood function [4]. Smart meter errors of harmonic could also be verified by testing devices or platforms [5,6]. However, as it takes a lot of manpower and time to dismantle the smart meters and bring them to a laboratory to calibrate, which also leads to the interruption of measurement, the methods based on remote calibration are of concern. The remote calibration could be realized by such as solving the parameter degradation network [7], installing remote calibration and monitoring systems, which consist of some monitoring devices, communication networks, and main station. Bayesian hierarchical model [8] and central limit theorem [9] usually be used in the remote monitoring management of the main station. The methods above require the installation of a large number of standard terminal devices, increasing investment, operation, and maintenance costs.

Due to the construction of Advanced Metering Infrastructure (AMI), many high-frequency and multi-type measurement data could be obtained, of which the application comes into focus. How to apply big data technology to realize the remote analysis of smart meter errors has also become a research hotspot. For example, Yip and Tan et al. [10] applied linear regression and linear

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Nomenclature		$\mathbf{R}'(t)$	Matrix of $\mathbf{r}'(t)$
<i>Abbreviations</i>		ξ_i	Correction of the i^{th} user meter
AMI	Advanced Metering Infrastructure	ξ	Matrix of ξ_i
MDMS	Meter Database Management System	x_i	Error of the i^{th} user meter
ADCS	Automatic Data Collection System	$\hat{x}_i$	Estimation of x_i
DREM	Dimension Reduction Estimation Model	$\mathbf{X}$	Matrix of x_i
KEM	Korhonen Estimation Model	$\mathbf{X}'(t)$	Extended matrix of $\mathbf{X}$
CNLREM	Constant Network Loss Rate Estimation Model	$\mathbf{X}'$	Estimation of $\mathbf{X}'$
CR	Current-Resistance	$\mathbf{X}'(t)$	$\mathbf{X}'$ in the t^{th} recursion
PL	Power-Loss	$\Delta w(t)$	Real value of network loss
DRLS	Damped Recursion Least Square	$\mathbf{w}_{\Delta}(t)$	Matrix of $\Delta w(t)$
<i>Indices</i>		$\Delta \hat{w}$	Estimation of $\Delta w(t)$
t	t^{th} measurement or recursion	$M(t)$	Membership sequence of t^{th} measurement
i	Number of user meters	$\mathbf{M}(t)$	Matrix of $M(t)$
$k(k = 1, 2, \dots, s)$	Number of elements in $\hat{\chi}$	$\varepsilon(t)$	Residual
<i>Parameters</i>		$\varepsilon(t)$	Matrix of $\varepsilon(t)$
T	Length of measurement period	$\mathbf{W}(t)$	Forgetting weight matrix
m	Total number of user meters	$r_{\text{ex}}(t)$	Extended measurement item
ρ	Forgetting factor	$\hat{r}_{\text{ex}}(t)$	Approximation of $r_{\text{ex}}(t)$
μ_0	Damping factor	$\mathbf{R}_{\text{ex}}(t)$	Matrix of $r_{\text{ex}}(t)$
s	Total number of clustering categories	$\hat{\mathbf{R}}_{\text{ex}}(t)$	Matrix of $\hat{r}_{\text{ex}}(t)$
$\mathbf{I}$	Unit matrix	$\chi(t)$	Real value of extended unknown
<i>Variables</i>		χ	Matrix of $\chi(t)$
$y(t)$	Active power of feeder meter	$\hat{\chi}_k$	Estimation of $\chi(t)$
$r_i(t)$	Reading value of i^{th} user meter	$\hat{\chi}$	Matrix of $\hat{\chi}$
$\mathbf{r}(t)$	Matrix of $r_i(t)$	$i(t)$	Current of feeder meter
$\mathbf{r}'(t)$	Extended matrix of $\mathbf{r}(t)$	$\Delta y(t)$	Total loss
$\mathbf{r}'(t)$	Approximation of $\mathbf{r}'(t)$	$\Delta \hat{y}(t)$	Approximation of $\Delta y(t)$
$\hat{\mathbf{Y}}_{\Delta}(t)$	Matrix of $\Delta \hat{y}(t)$	$\mathbf{Y}(t)$	Matrix of $y(t)$
$\bar{y}(t)$	Average of $\mathbf{Y}(t)$	$\mathbf{Y}_{\Delta}(t)$	Matrix of $\Delta y(t)$
$y^*(t)$	p.u. of $y(t)$	$\Delta \bar{w}(t)$	Estimation of $\Delta \bar{w}(t)$
$\Pi(t)$	Real product of Equation resistance and T	$\mu(t)$	Damping gain
$\hat{\Pi}_k$	Estimation of $\Pi(t)$	$J_{\mu}(t)$	Objective of DRLS
$\Delta \omega(t)$	Real value of network loss rate	$\mathbf{P}_{\mu}(t)$	Covariance matrix of DRLS
$\mathbf{R}(t)$	Matrix of $\mathbf{r}(t)$	$D_x(\%)$	RMSE of error estimation
$\mathbf{R}'(t)$	Matrix of $\mathbf{r}'(t)$	$D_{\text{loss}}(\%)$	RMSE of network loss rate estimation
		$C_d(\%)$	Rate of success of estimation
		$C_e(t)$	Criteria of success
		$\Delta \bar{w}(t)$	Real average of network loss rate

programming to build an anomaly monitoring framework, which could identify the defective meters and stealing electricity by analyzing measurement data. Mesbah [11] used linear error correction block coding to ensure that smart meter readings, which are not tampered with or out of order. Guo [3] proposed a remote estimation of smart meter errors based on conservation of energy and explained in detail, in which network loss is regarded as a dummy load. Many other estimation methods are also proposed, such as the method based on AMR and Recursion Least Squares (RLS) [12], k-means clustering [13], and the method based on Double-Parameter RLS [14]. However, although some research achievements have been made, there are still some shortcomings in accuracy, stability, and applicability of estimation in existing methods. Especially, the high calculation accuracy of network loss in advance is strictly required.

Network loss plays an important role in the remote estimation of smart meter errors based on big data, which seriously affects the accuracy of errors estimation results. When network loss changes drastically or frequently, the remote error estimations based on

traditional methods may fail or get wrong results [12]. Moreover, low voltage distribution networks usually have the characteristics of unknown topology and parameters, and low online measurement rate, which makes it difficult to obtain high accurate network loss. To solve this problem, some methods based on measurement data are proposed, such as the loss calculation method with distributed network impedance based on the big data obtained by smart grid [15] and so on. The Methods above aim to obtain the high accuracy value of network loss, which could reduce the effect of network loss on error estimation. The improvement of the estimation model and algorithm could also have the same effect. Least Squares and its improvement, such as Damped Recursion Least Squares (DRLS), partial least squares, weighted least squares, etc. [16–21] are widely used in engineering practice, in which DRLS prevents the phenomenon of parameter explosion, and has better convergence and stability [18].

On the base of the existing research [3,12,13,18], a remote estimation method of smart meter errors based on Dimension Reduction Estimation Model (DREM) and DRLS is proposed in the

paper, which realizes the estimation of network loss and smart meter errors at the same time. The main contributions of this paper are as follows :

- (1) Base on the dimension reduction process, the DREM is proposed, which makes the solution of model possible and realize the real-time estimation of network loss and smart meter errors without calculating network loss in advance;
- (2) DRLS is used to ensures the utilization of the new data and narrows the variation range of estimator, which has better robustness, and could reduce the effect of network loss;
- (3) Based on the analysis of the distribution feeder unit, a method to realize online remote estimation of large-scale smart meter errors is proposed, which obtains the variation of smart meters states, and online detection of power theft and leakage.

The structure of this paper is as follows: Section 2 introduces the smart meter error sources, data sources and ideas estimation for smart meter errors. Section 3 describes the traditional models of smart meter errors, and the corresponding online application process of remote estimation of smart meter errors based on DREM and DRLS is proposed. The effectiveness of the model and algorithm are verified in Section 4. Finally, Section 5 summarizes the result for

the proposed method.

2. Analysis and information acquisition of smart meter errors

2.1. Analysis of smart meter errors

The operating environment and structure of smart meters are shown in Fig. 1. The error sources mainly come from three aspects: (1) transformer errors; (2) meter internal errors, and (3) communication errors.

In distribution network, there are potential transformer and current transformer connecting to smart meter, where transformer errors may occur. The transformer errors are mainly from the hysteresis characteristics of current transformer, which causes the primary current cannot be replicated accurately into secondary current [22], especially in the case of light load ($I < 0.1I_N$) where harmonics takes the larger part of the load current. A smart meter usually consists of measurement unit, power module, storage module, sensor, communication module, microcontroller unit and so on. The meter internal errors mainly refer to dynamic measurement errors caused by the inadaptability of the accumulative time unit [23] and the deviation of the gain dynamic switching and power filter in microcontroller unit [24]. The measurement data collected by smart meters, such as active power, reactive power,

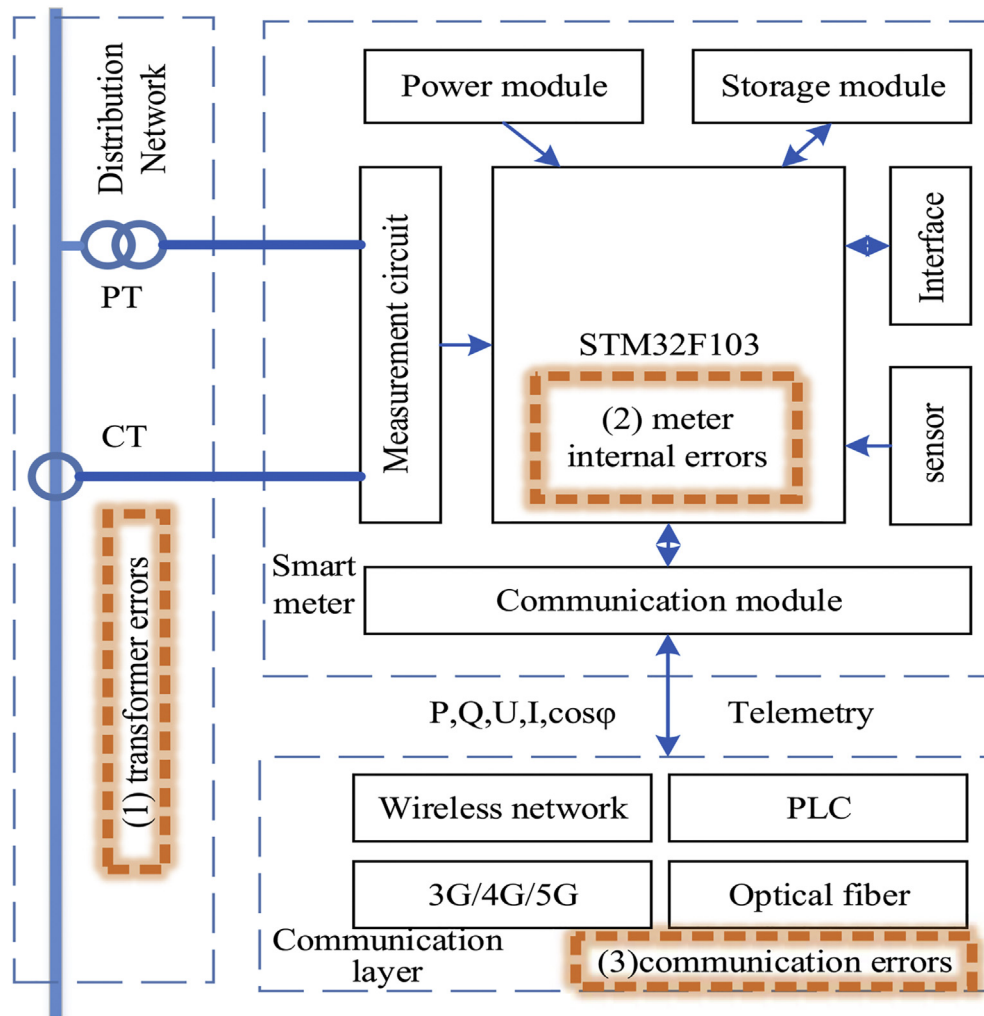


Fig. 1. The operating environment and structure of the smart meter.

voltage, etc. are transmitted to station through communication layer, which consists of wireless network, optical fiber and so on. The communication errors are mainly manifested as missing values and bad values, which are usually caused by PLC deterioration [25] or disturbance of surrounding wireless signals etc. [26]. Since the Meter Database Management System (MDMS) and Auto Data Collection System (ADCS) in station can manage the communication errors data before storage [27–29], the transformer errors and meter internal errors are taken into account and managed by technical means in the remote estimation of smart meter errors.

2.2. Information acquisition and the idea of smart meters 'errors remote estimation

Based on large-scale measurement data, the proposed remote estimation method for smart meter operating errors is mainly composed of three steps: (1) Obtain the information required by the estimation of smart meter errors and preprocess the acquired information to meet the requirements of the constructed estimation model; (2) Build remote estimation model of operating smart meter errors, then use the corresponding algorithm to solve the estimation model; (3) Check the accuracy of the estimation result. The process in detail is shown as Fig. 2., and the third step plays the most important role.

The information required by the remote estimation of smart meter errors includes static information and dynamic information. The static information is the information of distribution feeder unit and users, including (1) User name, number, address, power type, etc. (2) Smart meter information, such as smart meter number, name, type, address, etc. (3) Main electrical characteristic parameters related to the network topology of the distribution feeder unit, such as power supply radius, total length of low-voltage line, etc. (4) The main electrical characteristic parameters related to load, such as load rate, power nature and ratio, etc. [30].

The dynamic information is related to measurement data collected by the smart meter, such as electric quantity, power, voltage, current of feeder meter and user meters, etc. These data could be obtained by the smart meter data collection system of AMI, which includes smart meters, communication networks, and data management system. The energy and alert information are collected by smart meters, then be sent to the concentrator through PLC, BPL and so on [31]. The concentrator uploads the data to data center in main station through optical fiber, internet, WAN and so

on. The finishing touch is processing and storing the measurement data, which completed by MDMS and Auto Data Collection System (ADCS) [27].

3. Remote estimation model and solution algorithm of smart meter errors

3.1. Actual model of smart meters 'errors

A classic topology of the distribution feeder unit is shown as Fig. 3, including a feeder smart meter and m user smart meters. According to the conservation of energy, at any time period, the energy flowing into the area is equal to the sum of the energy flowing out and the network loss [3]. Thus, there is a relationship between the reading data for smart meters on a feeder:

$$y(t) = \mathbf{r}(t)\xi + \Delta w(t) \quad (1)$$

where $y(t)(\text{kWh})$ is the active energy reading value increment during the t^{th} measurement period of the feeder meter; $\mathbf{r}(t)_{1 \times m} = [r_1(t)r_2(t)\cdots r_m(t)]$ is the reading value matrix of user meters, consist of $r_i(t)(\text{kWh})$, which is the active energy reading value increment during the t^{th} measurement period of i^{th} user; $\xi_{m \times 1} = [\xi_1 \cdots \xi_i \cdots \xi_m]^T$ is the correction matrix of user meters, consist of $\xi_i(\%)$, which is the correction of the i^{th} user meter and assumed as a constant during the t^{th} measurement period; $\Delta w(t)(\text{kWh})$ is the network loss during the t^{th} measurement period in this distribution feeder unit.

Eq. (1) is the relationship of a single measurement. By arranging all measurement from 1^{th} to t^{th} , setting $x_i = \xi_i - 100\%$ as the errors of i^{th} meter, the multi-time scale measurement relationship could be obtained:

$$\mathbf{Y}_d(t)_{t \times 1} = \mathbf{R}'(t)_{t \times (m+t)} \mathbf{X}'(t)_{(m+t) \times 1} \quad (2)$$

where $\mathbf{Y}_d(t)_{t \times 1}$ is total loss sum caused by network loss and meter's errors; $\mathbf{R}'(t)_{t \times (m+t)}$ is the extended measurement matrix of user meters; $\mathbf{X}'(t)_{(m+t) \times 1}$ is the extended real errors matrix. These three variables are denoted as follows :

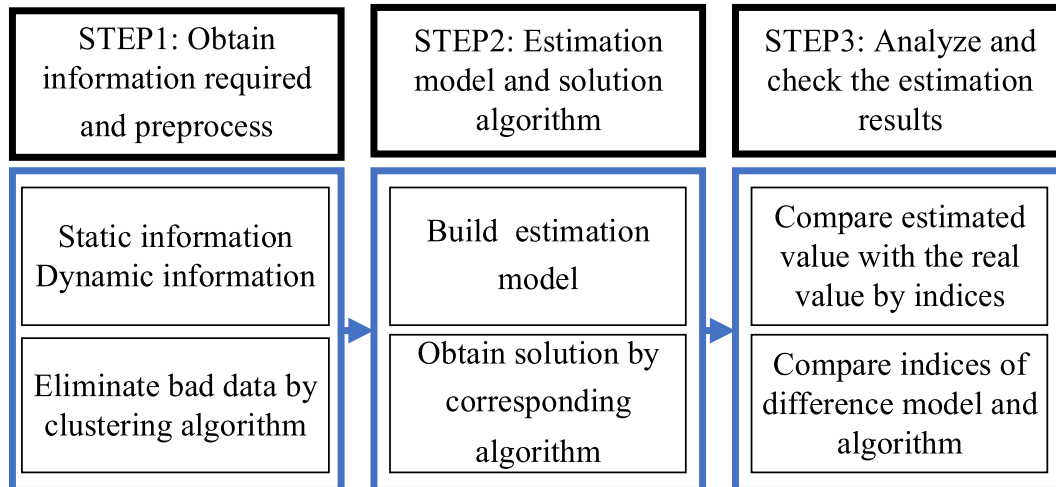


Fig. 2. Process of remote estimation of smart meter errors.

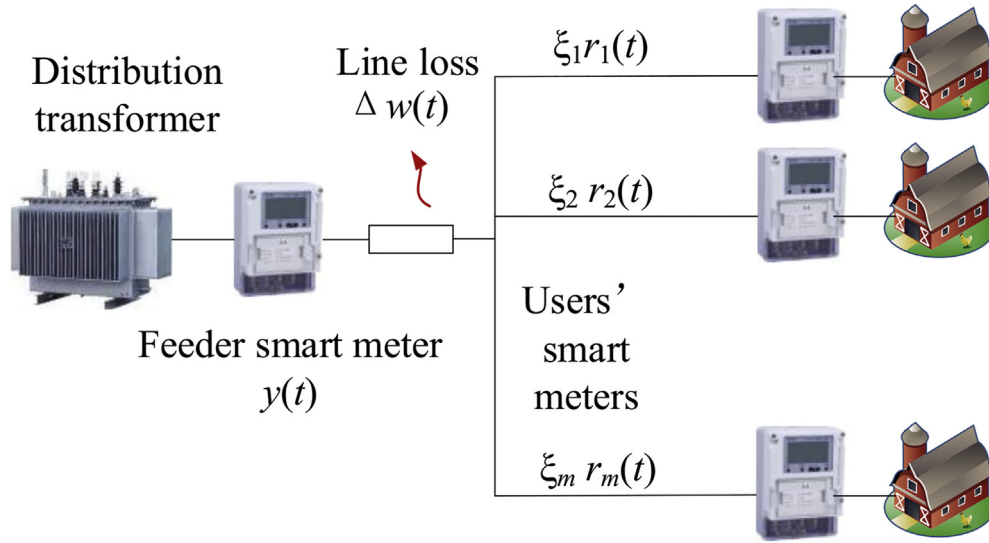


Fig. 3. Classic topology of the distribution feeder unit.

$$\mathbf{Y}_\Delta(t)_{t \times 1} = [\Delta y(1) \Delta y(2) \cdots \Delta y(t)]^T \quad (3)$$

$$\mathbf{R}'(t)_{t \times (m+t)} = [\mathbf{R}(t)_{t \times m} \mathbf{I}_{t \times t}] \quad (4)$$

$$\mathbf{X}'(t)_{(m+t) \times 1} = \begin{bmatrix} \mathbf{X}_{m \times 1} \\ \mathbf{w}_\Delta(t)_{t \times 1} \end{bmatrix} \quad (5)$$

where

$$\Delta y(t) = y(t) - \sum_{i=1}^m r_i(t) \quad (6)$$

$$\mathbf{R}(t)_{t \times m} = \begin{bmatrix} \mathbf{r}^{(1)}_{1 \times m} \\ \mathbf{r}^{(2)}_{1 \times m} \\ \vdots \\ \mathbf{r}^{(t)}_{1 \times m} \end{bmatrix} \quad (7)$$

$$\mathbf{X}_{m \times 1} = [x_1 x_2 \cdots x_m]^T \quad (8)$$

$$\mathbf{w}_\Delta(t)_{t \times 1} = [\Delta w(1) \Delta w(2) \cdots \Delta w(t)]^T \quad (9)$$

Eq. (2) is the actual model of smart meter errors, which is a set of linear Equations consist of t Equations, and could be obtained by t measurements. $\mathbf{X}'$ is a matrix of unknowns to solve for, containing $m + t$ unknowns. Because the number of unknowns is greater than that of Equations, this actual model of smart meter errors is unsolvable. Thus, the proper approximation should be taken to reduce the dimension of unknowns, making the number of unknowns is fewer than that of Equations.

3.2. Dimension Reduction Estimation Model of smart meter errors

The main idea of Dimension Reduction Estimation Model (DREM) of smart meters is approximately replacing the unknowns by clustering centers of unknowns, transferring the unsolvable actual model into a solvable approximation model.

Now, the variables in Eq. (2) are redefined as follows: $\mathbf{w}_\Delta(t)$, the extended unknown item in $\mathbf{X}'$, is set as $\chi(t) = [\chi(1), \chi(2), \cdots, \chi(t)]^T$, of which the dimension is $t \times 1$; Correspondingly, unit matrix $\mathbf{I}$, which is the extended measurement item in $\mathbf{R}'(t)$, is set as $\mathbf{R}_{\text{ex}}(t) =$

$\text{diag}(r_{\text{ex}}(1), r_{\text{ex}}(2), \cdots, r_{\text{ex}}(t))$, of which the dimension is $t \times t$.

The dimension reduction process is shown as Fig. 4. The 7 circles on the black solid line represent 7 actual network losses (i.e. unknowns), which are indicated by $\chi(k) (k = 1, 2, \cdots, 7)$. It can be seen that network losses are frequent and change a lot. According to actual model, we have to solve these 7 unknowns one by one, and this solution has been proved impossible. While in the Dimension Reduction Estimation Model (DREM), the 7 network losses are clustered into 3 classes (i.e., $s = 3$), and the clustering result is indicated by membership matrix $\mathbf{M}(t)$. The red marks on the red dotted line represent the clustering centers of the classes $\hat{\chi}_n (n = 1, 2, 3)$. The square, triangle and cross separately indicate. Each circle belongs to only one certain clustering center. According to membership matrix $\mathbf{M}(t)$, $\chi(2), \chi(4), \chi(6)$ belong to $\hat{\chi}_2$; $\chi(3), \chi(7)$ belong to $\hat{\chi}_1$ while $\chi(1), \chi(5)$ belong to $\hat{\chi}_3$. The main idea of DREM is approximately replacing the unknowns by clustering centers of unknowns, transferring the unsolvable actual model into a solvable approximation model. For example, replace $\chi(2), \chi(4), \chi(6)$ by $\hat{\chi}_2$, replace $\chi(1), \chi(5)$ by $\hat{\chi}_3$, and the approximation result is shown as red dotted line in Fig. 4. This approximation reduces the number of unknowns on the basis of basically restoring the variation trend of

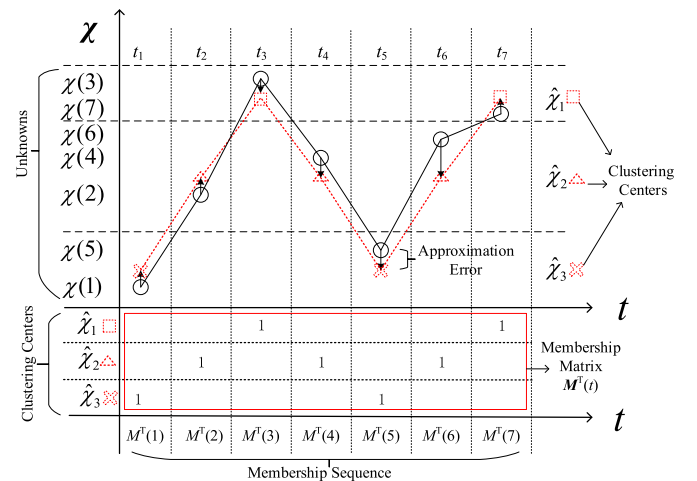


Fig. 4. The principle of dimension reduction estimation model.

unknowns. The differences between the actual model and DREM mainly lie in the approximation errors, which is representatively marked in Fig. 4. It is obvious that even if the network losses are frequent and change a lot, there is still one certain clustering center to carried on replacement and approximation. it is also obvious that when $s \rightarrow t$, $\hat{\chi} \rightarrow \chi(t)$. The membership matrix $\mathbf{M}(t)$, of which dimension is $t \times s$, shown below Fig. 4, denotes the subordination of each $\chi(t)$ to the clustering centers. The $\mathbf{M}(t)$ is explained as follows, in which $M(t)$ is the membership sequence of t^{th} measurement:

$$\mathbf{M}(t)_{t \times s} = \begin{bmatrix} M(1)_{1 \times s} \\ M(2)_{1 \times s} \\ \vdots \\ M(t)_{1 \times s} \end{bmatrix} \quad (10)$$

However, $\chi(t)$ is unknown, it is impossible to obtain actual $\mathbf{M}(t)$ by clustering $\chi(t)$. If the selection of $\mathbf{R}_{\text{ex}}(t)$ and $\hat{\chi}$ is suitable, the $\mathbf{M}(t)$ obtained by clustering $\mathbf{R}_{\text{ex}}(t)$ is almost the same. Thus, $\mathbf{M}(t)$ could be obtained by clustering $\mathbf{R}_{\text{ex}}(t)$. The more detailed schematic diagram of dimension reduction is shown in Appendix A.

Mr. Korhonen [12] set $s = 1$, $r_{\text{ex}}(t)_{1 \times 1} = 1$ and $\hat{\chi} = \Delta \hat{w}$ to

estimate the meter errors. However, the results are shown in Fig. 5, which could barely distinguish the out of tolerance meter 10, shown as red dotted line. Moreover, with the change of network loss, the estimation values of meter errors are unstable. The reason is that the selection of $\mathbf{R}_{\text{ex}}(t)$ and $\hat{\chi}$ is unsuitable and s is too small, which causes $\hat{\chi}$ not similar enough to $\chi(t)$ and further affects all estimation values.

To select $\mathbf{R}_{\text{ex}}(t)$ and $\hat{\chi}$ properly, the equivalent resistance method is applied, which is widely used in calculating network loss [32]. Two feasible options are proposed in the paper. The first option is Current-Resistance (CR), which means the extended measurement item is current and the extended unknown item is equivalent resistance; the second option is Power-Loss (PL), which means the extended measurement item is power while the extended unknown item is network loss rate. The specific derivation of the options is shown in Appendix B. The details of the options are as follows:

Option CR:

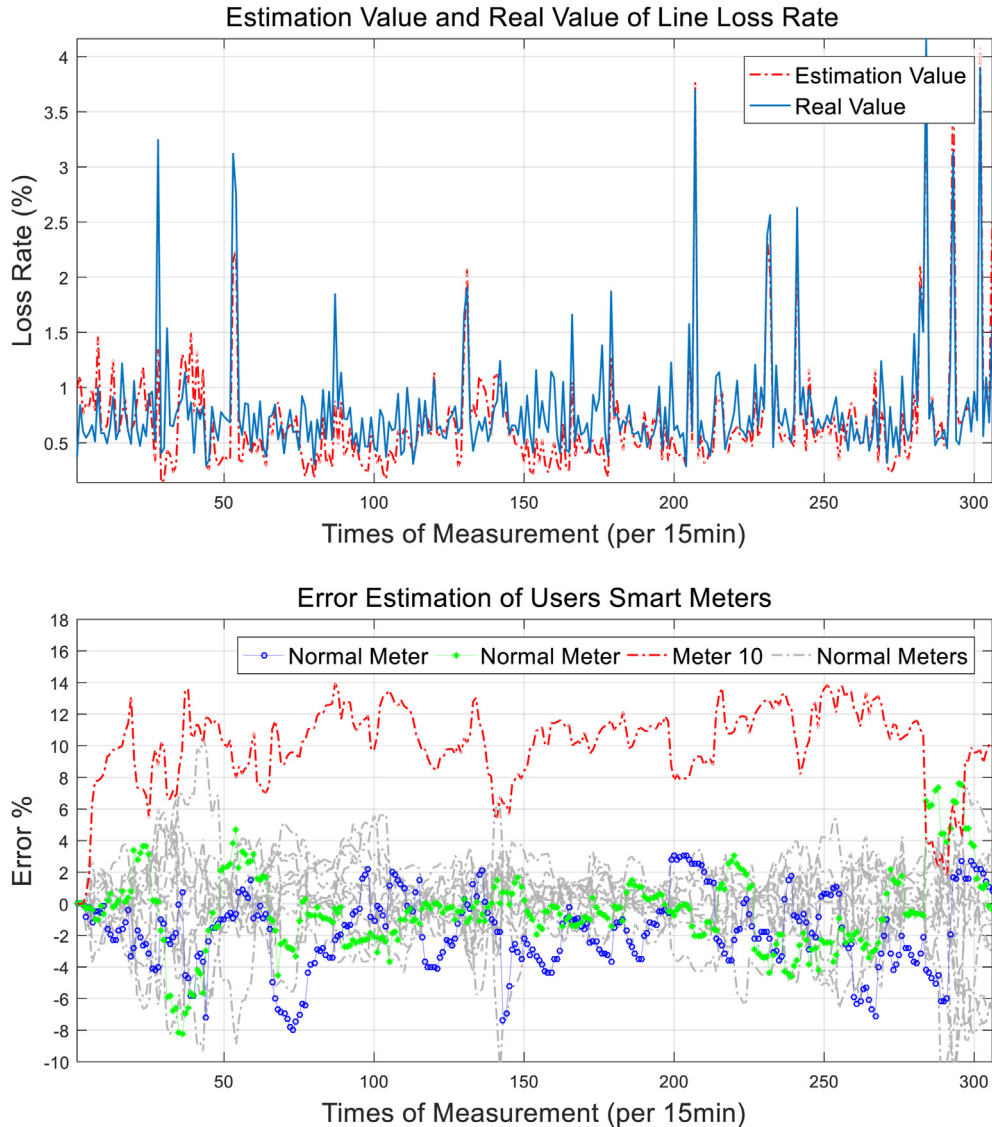


Fig. 5. The errors estimation result of Korhonen estimation model.

$$r_{\text{ex}}(t) = i^2(t) \quad (11)$$

$$\hat{\chi}_{s \times 1} = [\hat{\Pi}_1 \hat{\Pi}_2 \cdots \hat{\Pi}_s]^T \quad (12)$$

where $i^2(t)$ (AA) is the square of the feeder meter current during the t_{th} measurement period; $\hat{\Pi}_k (k = 1, 2, \dots, s)$ is the estimation value of the product of the equivalent resistance and the length of the measurement period, which is clustered by $i^2(t)$.

Option PL:

$$r_{\text{ex}}(t) = y^*(t)y(t) \quad (13)$$

$$y^*(t) = y(t)/\bar{y}(t) \quad (14)$$

$$\bar{y}(t) = \sum_{i=1}^t y(i)/t \quad (15)$$

$$\hat{\chi}_{s \times 1} = [\Delta\bar{\omega}_1 \Delta\bar{\omega}_2 \cdots \Delta\bar{\omega}_s]^T \quad (16)$$

where $y^*(t)$ is the p. u. value of $y(t)$; $\bar{y}(t)$ is the average value of $y(t)$; $\Delta\bar{\omega}(\%)$ is the estimation value of the average value of the network loss rate.

The advantages of Option CR are as follows:

- (1) The extended estimation item Π depends on the network parameters of the distribution feeder unit, which is relatively stable compared with the drastic-varying network loss. Thus, the estimation process is smoother and estimation values are easy to converge to real values and more accurate.
- (2) The measurement data types involved are only $i(t)$ more than those in the actual model, and $i(t)$ is easy to obtain.

However, the current is not required in most marketing system at present, so $i^2(t)$ could be hardly obtained. In this case, The Option PL could be applied. Option PL only requires $y(t)$ and $\mathbf{r}(t)$, which could be easily obtained by current measurement systems.

After the selection of $\mathbf{R}_{\text{ex}}(t)$ and $\hat{\chi}$, the $\mathbf{M}(t)$ be obtained by the k-means cluster algorithm. Then, the Dimension Reduction Estimation Model (DREM) could be built as follows:

Single measurement:

$$\hat{y}_{\Delta}(t) = \mathbf{r}'(t)_{1 \times (m+s)} \hat{\mathbf{X}}(t)_{(m+s) \times 1} \quad (17)$$

Multi-time scale measurement:

$$\hat{\mathbf{Y}}_{\Delta}(t)_{t \times 1} = \hat{\mathbf{R}}(t)_{t \times (m+s)} \hat{\mathbf{X}}(t)_{(m+s) \times 1} \quad (18)$$

where

$$\hat{\mathbf{r}}(t)_{t \times (m+s)} = [\mathbf{r}(t)_{1 \times m} \hat{\mathbf{r}}_{\text{ex}}(t)_{1 \times s}] \quad (19)$$

$$\hat{\mathbf{R}}(t)_{t \times (m+s)} = [\mathbf{R}(t)_{t \times m} \hat{\mathbf{R}}_{\text{ex}}(t)_{t \times s}] \quad (20)$$

$$\hat{\mathbf{X}}_{(m+s) \times 1} = [(\hat{\mathbf{X}}_{m \times 1})^T (\hat{\chi}_{s \times 1})^T]^T \quad (21)$$

$$\hat{\mathbf{R}}_{\text{ex}}(t)_{t \times s} = \begin{bmatrix} \hat{\mathbf{r}}_{\text{ex}}(1)_{1 \times s} \\ \hat{\mathbf{r}}_{\text{ex}}(2)_{1 \times s} \\ \vdots \\ \hat{\mathbf{r}}_{\text{ex}}(t)_{1 \times s} \end{bmatrix} \quad (22)$$

$$\hat{\mathbf{r}}_{\text{ex}}(t)_{1 \times s} = r_{\text{ex}}(t) \mathbf{M}(t)_{1 \times s} \quad (23)$$

$$\hat{\chi}_{s \times 1} = [\hat{\chi}_1 \hat{\chi}_2 \cdots \hat{\chi}_s]^T \quad (24)$$

And the estimation value of network loss during the t_{th} measurement period is:

$$\Delta\hat{\omega}(t) = (\hat{\mathbf{r}}_{\text{ex}}(t) \hat{\chi}) / y(t) \quad (25)$$

3.3. Damped Recursion Least Squares

The Recursion Least Squares (RLS) is used to solve the estimation model. However, there are some problems, such as parameter explosion, identification process ending before converging when the initials are unsuitable, instability caused by random noise [18]. The damping item is introduced into RLS, which limits the variation range of estimation values. The objective function of Damped Recursion Least Squares (DRLS) based on DREM algorithm is as follows:

$$J_{\mu}(t) = \varepsilon(t)^T \mathbf{W}(t) \varepsilon(t) + \mu(t) [\hat{\mathbf{X}}(t) - \hat{\mathbf{X}}(t-1)]^T [\hat{\mathbf{X}}(t) - \hat{\mathbf{X}}(t-1)] \quad (26)$$

where $\varepsilon(t)$ is residual:

$$\varepsilon(t) = \mathbf{Y}_{\Delta}(t) - \hat{\mathbf{Y}}_{\Delta}(t) \quad (27)$$

$\mathbf{W}(t)$ is forgetting weight matrix during the t_{th} measurement period:

$$\mathbf{W}(t) = \begin{bmatrix} \rho \mathbf{W}(t-1) & 0 \\ 0 & 1 \end{bmatrix} \quad (28)$$

where ρ is the forgetting factor, and $0 < \rho < 1$. Usually, new measurement data with higher weight. The weight of the newest measurement data be 1, and the weight of old data be exponential decay. The value ρ reflects the influence of old data on the estimation of meter errors. The larger of ρ , and the slower the variation of estimation value would be, which means of low instantaneity; Conversely, if ρ is small, the old data would be soon forgotten, and the instantaneity would be raised. Mr. Korhonen has further discussed ρ in other respects [12].

$\mu(t)$ is damping gain:

$$\mu(t) = \mu_0 \times \text{mean}(\mathbf{r}'(t)^T \mathbf{r}'(t)) \quad (29)$$

where μ_0 denotes damping factor, generally set as 0.01, mean is the average value of all elements in a matrix.

By minimizing Eq. (26), the Equation of estimation values could be obtained, the specific derivation is shown in Appendix C:

$$\begin{aligned} \partial J(t)/\partial \hat{\mathbf{X}}(t) &= 0 \\ \Downarrow \\ \hat{\mathbf{X}}(t) &= \hat{\mathbf{X}}(t-1) + \rho\mu(t)\mathbf{P}_\mu(t)[\hat{\mathbf{X}}(t-1) - \hat{\mathbf{X}}(t-2)] + \mathbf{P}_\mu(t)\mathbf{r}'(t)^T[\Delta y(t) - \mathbf{r}'(t)\hat{\mathbf{X}}(t-1)] \end{aligned} \quad (30)$$

where $\mathbf{P}_\mu(t)$ is covariance matrix, of which dimension is $(m+s) \times (m+s)$:

$$\mathbf{P}_\mu(t) = [\mu(t)\mathbf{I} - \rho\mu(t)\mathbf{I} + \rho\mathbf{P}_\mu(t-1)^{-1} + \mathbf{r}'(t)^T\mathbf{r}'(t)]^{-1} \quad (31)$$

Eq. (30) and Eq. (31) are the recursion formulas for the remote estimation method of smart meter errors based on DREM and DRLS.

3.4. Online application process for smart meter errors estimation based on DREM and DRLS

The online application process for smart meter error estimation based on DREM and DRLS mainly including three parts of algorithms: initialization, DREM building, DRLS solution.

Algorithm 1 Initialization.

Input: $\mathbf{Y}(t)$, $\mathbf{i}^2(t)$, s
Output: $\mathbf{R}_{\text{ex}}(t)$, $\mathbf{P}_\mu(1)$, $\hat{\mathbf{X}}'(1)$, $\hat{\mathbf{X}}'(2)$
1: if $\mathbf{i}^2(t)$ is accessible
2: $\mathbf{R}_{\text{ex}}(t) = \mathbf{i}^2(t); \hat{\chi}_{s \times 1} = [\hat{\Pi}_1 \hat{\Pi}_2 \dots \hat{\Pi}_s]^T$
3: else
4: $\mathbf{R}_{\text{ex}}(t) = \mathbf{Y}^*(t) \cdot \mathbf{Y}(t); \hat{\chi}_{s \times 1} = [\Delta \hat{\omega}_1 \Delta \hat{\omega}_2 \dots \Delta \hat{\omega}_s]^T$
5: end
6: Initialize covariance matrix by $\mathbf{P}_\mu(1) = (0.001 \times \mathbf{I}_{(m+s) \times (m+s)})^{-1}$;
7: Initialize estimation value matrix by $\hat{\mathbf{X}}'(1:2) = [0_{m \times 1}^T; \hat{\chi}_{s \times 1}^T]^T$;

In Algorithm 1, a certain size of initial data is input, which obtained from the MDMS. According to the data type, the corresponding option is selected (line 1–4) and the extended measurement matrix $\mathbf{R}_{\text{ex}}(t)$ will be the input of DREM building. The total number of clustering categories s is set. The larger the s , the closer the estimation value $\hat{\chi}$ to extended unknown matrix χ , and more accurate the network loss and error estimation would be, while the cost is increasing the number of unknowns and the amount of computation. Otherwise, though the estimation would be less accurate with small s , the computing speed would be improved. The covariance matrix $\mathbf{P}_\mu(t)$ and estimation value matrix $\hat{\mathbf{X}}(t)$ are initialized and as the input of DRLS solution. The initial of $\mathbf{P}_\mu(t)$ should be a large enough invertible matrix [12]. The initial of meter error estimation value matrix $\hat{\mathbf{X}}(t)$ should be a zero matrix because the meter errors are usually around zero. As for the initial of extended unknown matrix $\hat{\chi}$, the multiplier should be taken into account.

Algorithm 2 DREM building.

Input: $y(t)$, $r_i(t)$, $r_{\text{ex}}(t)$
Output: $M(t)$, $y_\Delta(t)$, $\hat{\mathbf{r}}'(t)$, $\mathbf{R}_{\text{ex}}(t)$
1: Update extended measurement matrix by $\mathbf{R}_{\text{ex}}(t) = [\mathbf{R}_{\text{ex}}(t-1)^T; r_{\text{ex}}(t)]^T$

Algorithm 2 (continued)

2: Obtain membership sequence by $M(t) = k - \text{means}(\mathbf{R}_{\text{ex}}(t)^T)$
3: Obtain approximation of $r_{\text{ex}}(t)$ by $\hat{\mathbf{r}}_{\text{ex}}(t)_{1 \times s} = r_{\text{ex}}(t)M(t)_{1 \times s}$
4: Obtain total loss by $\Delta y(t) = y(t) - \sum_{i=1}^m r_i(t)$
5: Obtain approximation of $\mathbf{r}'(t)$ by $\hat{\mathbf{r}}'(t)_{1 \times (m+s)} = [\mathbf{r}(t)_{1 \times m}; \hat{\mathbf{r}}_{\text{ex}}(t)_{1 \times s}]$

In Algorithm 2, the newest t^{th} measurement data is input. The membership sequence $M(t)$ of new input data is obtained by clustering the new $\mathbf{R}_{\text{ex}}(t)$ (line 1–2). It's worth noting that because the clustering centers of k-means method are randomly selected, the membership sequence may be different in each time of clustering. Then the t^{th} total loss and approximation of t^{th} extended measurement matrix are obtained. Both of them will be the input of DRLS solution.

Algorithm 3 DRLS solution.

Input: $\mathbf{P}_\mu(t-1)$, $\hat{\mathbf{X}}'(t-1)$, $\hat{\mathbf{X}}'(t-2)$, $\Delta y(t)$, $\hat{\mathbf{r}}'(t)$, ρ , μ_0
Output: $\mathbf{P}_\mu(t)$, $\hat{\mathbf{X}}'(t)$, $\mu(t)$
1: Obtain damped gain by $\mu(t) = \mu_0 \times \text{mean}(\hat{\mathbf{r}}'(t)^T \hat{\mathbf{r}}'(t))$
2: Update covariance matrix by $\mathbf{P}_\mu(t) = [\mu(t)\mathbf{I} - \rho\mu(t)\mathbf{I} + \rho\mathbf{P}_\mu(t-1)^{-1} + \hat{\mathbf{r}}'(t)^T \hat{\mathbf{r}}'(t)]^{-1}$
3: Obtain estimation values by $\hat{\mathbf{X}}'(t) = \hat{\mathbf{X}}'(t-1) + \rho\mu(t)\mathbf{P}_\mu(t)[\hat{\mathbf{X}}'(t-1) - \hat{\mathbf{X}}'(t-2)] + \mathbf{P}_\mu(t)\hat{\mathbf{r}}'(t)^T[\Delta y(t) - \hat{\mathbf{r}}'(t)\hat{\mathbf{X}}'(t-1)]$
4: Plot the estimation values of user meter error by $\hat{\mathbf{X}}(t) = \hat{\mathbf{X}}'(1:m, t)$
5: Plot the estimation values of network loss rate by $\Delta \hat{\omega}(t) = (\hat{\mathbf{r}}_{\text{ex}}(t)\hat{\chi})/y(t)$
6: Go to Algorithm 2

In Algorithm 3, because two-step algorithm is involved, the $(t-1)^{\text{th}}$ and $(t-2)^{\text{th}}$ estimation matrix should be registered as input. The forgetting factor ρ is set to ensure the utilization of new data. The larger of ρ , the slower the variation of estimation value will be, which means of low instantaneity; Conversely, if ρ is small, the effects of old data decay soon, and the instantaneity will rise. The damped factor μ_0 is set to adjust the variation range of estimation matrix. If the μ_0 is zero, the DRLS degrades into RLS. The smaller μ_0 , the more sensitive to disturbance the estimation will be. While with large μ_0 , the variation range will narrow while estimation will be insensitive to the changing, which means more recursion (or measurements) is required for converging. To increase the adaptability of the algorithm, the damped gain $\mu(t)$ is corrected according to the newest extended measurement matrix (line 1). The estimation values of meter errors and network loss rate are extracted from the estimation matrix (line 4–5). Then the program returns to the part of DREM building and waiting for the next new data input.

The total flow of the online application process is shown as Fig. 6. By updating measurement data continuously, the errors of the user smart meters and the network loss rate could be monitored in real-time.

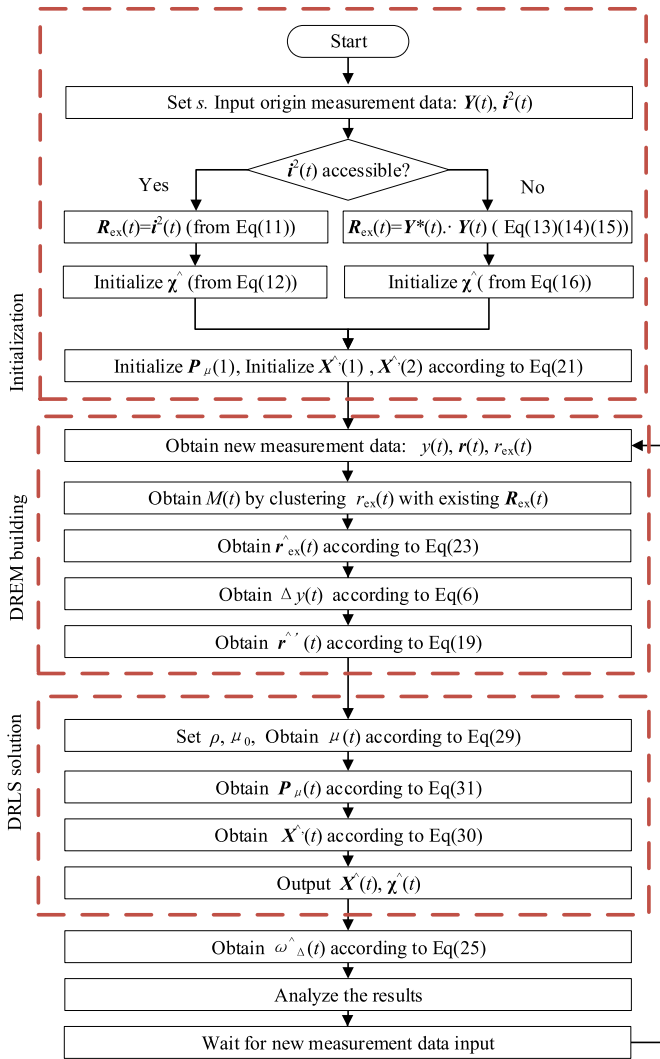


Fig. 6. The online application process for smart meter errors estimation based on DREM and DRLS.

4. Case study

To verify the proposed method, the experiments were carried out in laboratory and actual distribution feeder unit. In the laboratory case, the proposed method was compared with other methods and analyzed in detail. In the case of actual distribution feeder unit, the practicability of the proposed method was proved. Then on the base of actual case, the effects of the parameters involved in the proposed method were discussed. And the recommendations of parameter selection were proposed.

4.1. Analysis and validation of validity

For the convenience of analysis, the measurement data was obtained by a simulation distribution feeder unit in laboratory, which was controllable and observable. There were 15 smart meters being estimated. The meter No.10 was set as out of tolerance meter, of which error was 10% while there were no errors in the rest meters. The data collection frequency is every 15min; 305 times of measurement were recorded as input data; The data types include the real value of active power, current from a feeder meter, and the real value and the reading value of active power from 15 user

meters.

The methods with DREM, Korhonen Estimation Model (KEM) and Constant Network Loss Rate Estimation Model (CNLREM) were analyzed in the case studies. And the results of KEM was shown as Fig. 5. Moreover, the DRLS (proposed algorithm $\mu_0 = 0.01$) and RLS ($\mu_0 = 0$) algorithms were also analyzed. The k-means algorithm in MATLAB was used as cluster method. The parameter settings were $s = 20$; $\rho = 0.5$.

4.1.1. DREM with option CR (DREM-CR)

- (i) RL: The estimation results obtained by DREM-CR and RLS is shown as Fig. 7. There was deviation in the estimation results of network loss rate to some degree in this method. When the real values of network loss rate changed abruptly, the estimation values also varied greatly. The variation range of the estimation values of meter errors in this method is from -6% to $+4\%$. Therefore, DREM could estimate the network loss and meter errors in real time, but the estimation was sensitive to the variation of network loss, which means low stability and robustness.
- (ii) DRLS (proposed algorithm): The estimation results obtained by DREM-CR and DRLS is shown as Fig. 8. In terms of the process of the estimation, the estimation values of meter errors were hardly influenced by the abruptness of network loss rate. And the variation range of estimation value of errors was limited to a range from -2% to $+2\%$, which was further smaller than that of RLS. In terms of the estimation results, the estimation of network loss rate obtained by DRLS is more accurate than that obtained by RLS. The estimation errors of meter No.10 steadied around 10%, and the rest of meter estimation errors steadied around 0%, which means the meter out of tolerance was identified successfully. Therefore, the DRLS could effectively reduce the variation range of the estimation value and increase the stability, robustness and accuracy of the estimation.

4.1.2. DREM with option PL (DREM-PL)

- (i) RLS: The estimation results obtained by DREM-PL and RLS is shown as Fig. 9. Compared with DREM-CR, the estimation accuracy of both network loss and meter errors of DREM-PL was lower. And the variation range was larger, which ranged from -8% to $+4\%$. The reason was that DREM-PL ignored the reactive power and the variation of voltage.
- (ii) DRLS (proposed algorithm): The estimation results obtained by DREM-PL and DRLS is shown as Fig. 10. Compared with RLS, the estimation accuracy of DRLS is higher, and the variation range is smaller, which also proved the superior of DRLS. On the other hand, although the results of DREM-PL were not as good as DREM-CR, it was still acceptable, which verified the feasibility of the approximant.

4.1.3. CNLREM

CNLREM considers the network loss as the product of a certain constant network loss rate and $y(t)$ rather than an unknown varying with time. The constant network loss rate is supposed to be the average value, which could be approximated from statistic data. Thus, the number of unknowns would be m and dimension reduction would be unnecessary, which means the estimation model would be simple and easy to realize. However, the difference between the given value and the real value of the network loss would affect the estimated value of the meter errors. Thus, in the case of

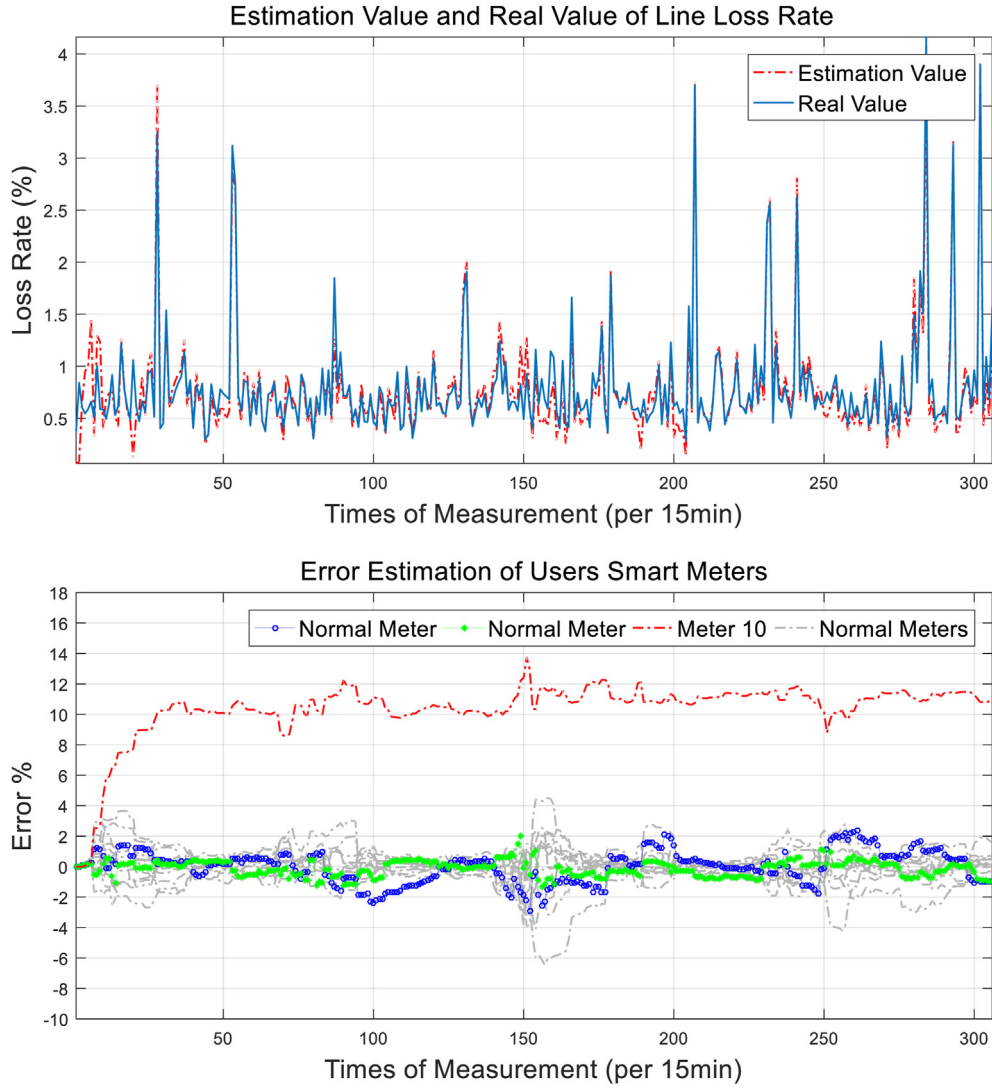


Fig. 7. The estimation results of DREM-CR and RLS.

network loss varying frequently and greatly, the estimation results (shown as Appendix D) were not good enough.

4.1.4. Index comparison

In order to compare the methods mentioned above more directly. Three indices are proposed as follows:

$$D_x = \sqrt{\sum_{i=1}^t \sum_{j=1}^m |\hat{x}_j(t) - x_j|^2 / t \times m(\%)} \quad (32)$$

$$D_{\text{loss}} = \sqrt{\sum_{i=1}^t |\Delta \hat{\omega}(t) - \Delta \omega(t)|^2 / t(\%)} \quad (33)$$

$$C_d = \left(\sum_{j=1}^t C_e(j) \right) / t(\%) \quad (34)$$

$$C_e(i) = \begin{cases} 1, & \text{if } \max |\hat{\mathbf{X}} - \mathbf{X}| < 0.01 \\ 0, & \text{else} \end{cases} \quad (35)$$

where D_x denotes Root Mean Squared Errors (RMSE) of meter error estimation, which explains the accuracy of meter error estimation. The method with low D_x deserve higher estimation accurate on meter errors; D_{loss} denotes RMSE of network loss rate estimation, and $\Delta \omega(t)$ is the real value of network loss rate, which is accessible in laboratory. Also, the method with low D_{loss} has higher estimation accuracy on network loss rate; C_d is the success rate of estimation, $C_e(i)$ is the criteria of successfully identifying the out of tolerance meters. When all of the differences between real value and estimation value of meter errors are smaller than 1%, the estimation in this measurement period is judged as success.

The index calculating results of the methods are shown in Table 1 it is known by comparing the columns that the RMSEs of the proposed algorithm are all lower than RLS, and the success rates are all higher than RLS, which verifies that the DRLS (proposed algorithm) could reduce the deviation degree of the estimation value from the real value, and improve the stability, robustness and accuracy of the estimation. By comparing the rows, it is seen that

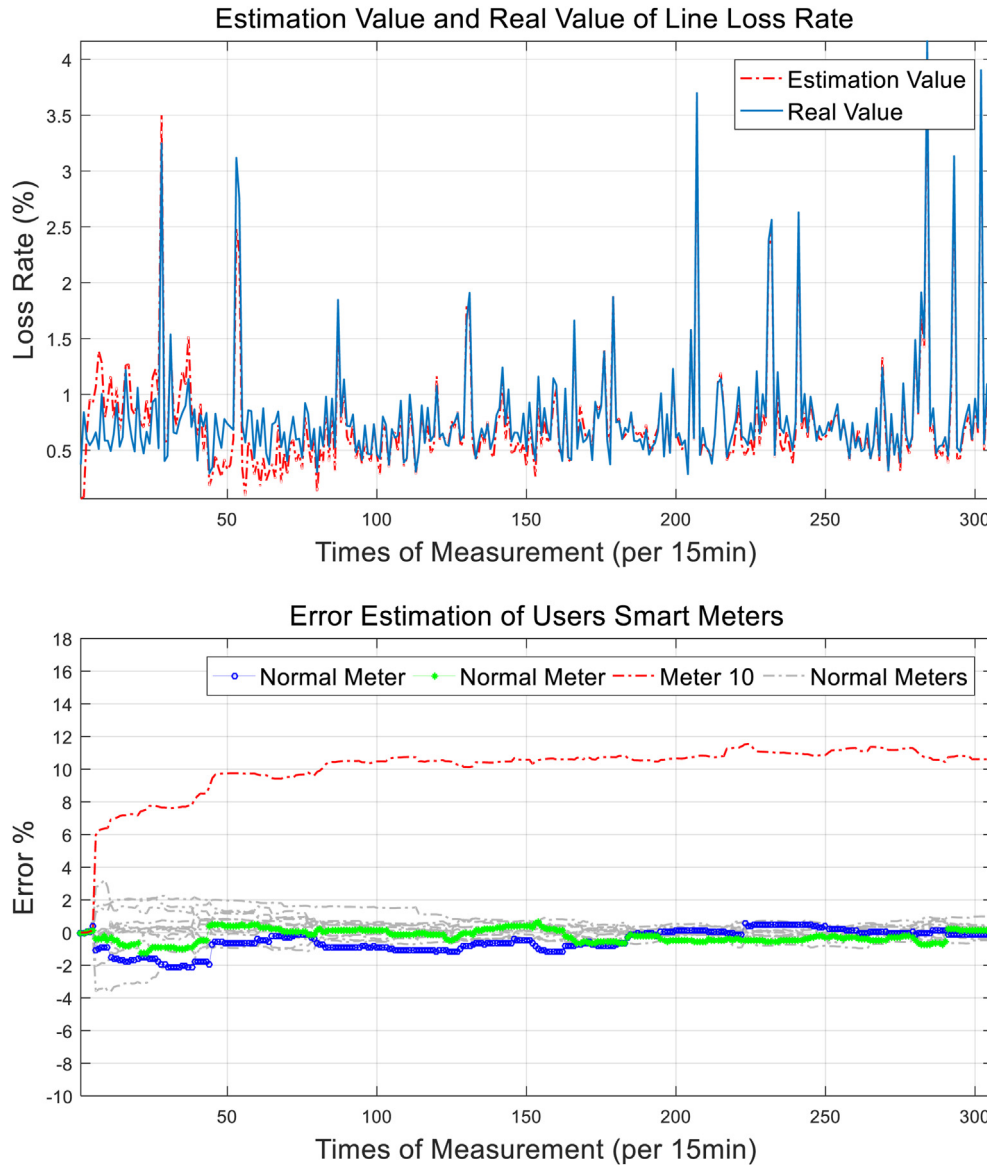


Fig. 8. The estimation results of proposed method (CR).

indices of the proposed models are optimal, those of KEM are second, and those of CNLREM are last. In the proposed method, the indices of DREM-CR and DREM-PL are closed to each other, while the former is superior to some degree. Thus, in the case that the current of a feeder meter is accessible, DREM-CR is recommended, otherwise the DREM-PL.

4.2. The application in actual distribution feeder unit

To verify the practicability, the proposed method was applied to estimate the errors of 112 smart meters in a distribution feeder unit in Tianjin, China. The data collection frequency is every 15min; 2880 times measurement are recorded; The data types include the real value of active power, current from a feeder meter, and the reading value of active power from 112 user meters. The applied models include DREM-CR (proposed model), KEM and CNLREM. The applied algorithm of all models was DRLS, of which the parameters were set as $\rho = 0.5$; $\mu_0 = 0.01$; $s = 20$. The estimation results of the proposed method are shown as Fig. 11. To make the

figure clear, only the first 1000 measurements are plotted. And the results of other methods are shown in Appendix E. The index calculating results are shown in Table 2. If the meters with estimated absolute values of more than 2% were considered to be the meters out of tolerance, the meters out of tolerance were No.59 and No.82. according to Fig. 11. The final estimation results of meter errors were obtained by averaging all estimation results in all measurement periods. And the estimation results of No.59 and No.82 were +5.00% and +9.10%. According to the on-site calibration by standard meters, the meters out of tolerance were No.14, No.59, No.82, of which the real errors were -1.84%, +4.61%, +9.22%. Because the error of No.14 was small, it was difficult to identify. In order to avoid omission of the meters with errors, it was suggested that the meters with the absolute value of the error estimation values exceeding 1% should be identified as out of tolerance. And the rates of omission and undue-detection (which means identifying the meter without error as the meter out of tolerance) are shown as Table 3.

Because the real values of network loss rate is inaccessible, there

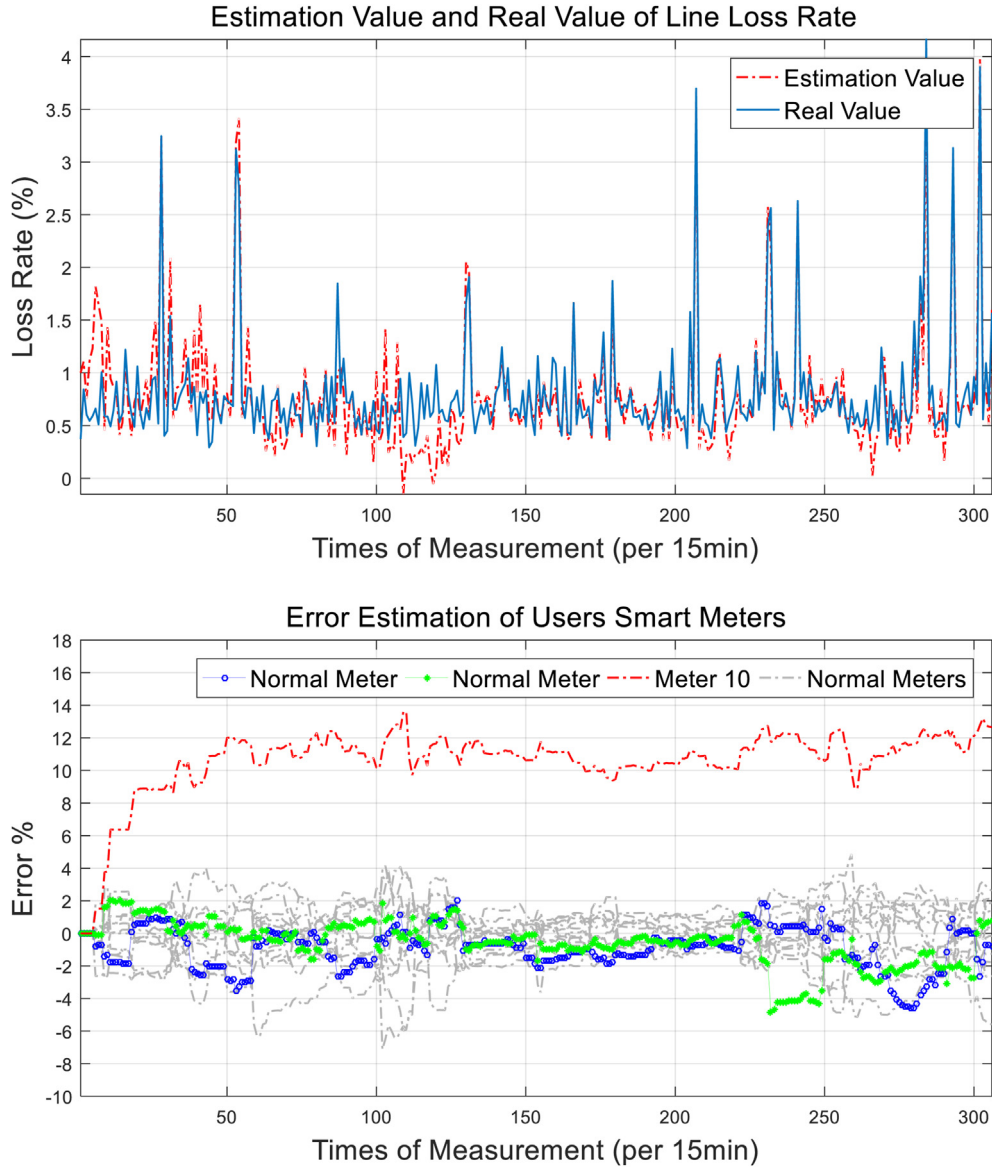


Fig. 9. The estimation results of DREM-PL and RLS.

was not $D_{\text{loss}}(\%)$ in Table 2. According to the indices, though Korhonen method could identify the meters with errors to some degree in the laboratory case, its performance was unsatisfactory in the actual case, let alone the Constant network loss rate method. In terms of rates of omission and undue-detection, since the threshold value is conservative, the rates of omission of all methods were 0%, while the rate of undue-detection of the proposed method was lower than other methods. Therefore, if applying the proposed method to locate the meter out of tolerance in advance, the workload of inspectors could be greatly reduced.

4.3. The further discussion on the proposed method

To further study the characteristics of the proposed and determine the optimal parameter selection, the membership matrix $\mathbf{M}(t)$ access methods, the effects of total number of clustering categories s , damped factor μ_0 and load condition are discussed based on the data from actual distribution feeder unit.

4.3.1. Parameter optimization

In the proposed method, three key parameters have great influence on the error estimation results, which are $\mathbf{M}(t)$, s and μ_0 . In order to determine the optimal parameter selection, an orthogonal experiment is used to study the effect of the combined effect of three parameters. The design of factors and levels is shown Table 4.

According to the study experience, the estimation results are better when the extended measurement item $\mathbf{R}_{\text{ex}}(t)$ is clustered to obtain the membership matrix $\mathbf{M}(t)$. Therefore, more detailed experiments should be conducted on this level. Therefore, the level 1 of factor A is repeatedly set to the level 5 to avoid the use of mixed orthogonal tables and simplify the experiment. According to Table 4, the selected orthogonal experiment [33] table is $L_{25}(5^6)$, which refers to 25 times tests including 5 levels and factors. Since the main concern in engineering application is the success rate of estimation, the success rate of estimation is selected as the experimental results for analysis. The results of 25 experiments are shown in Table 5. Mean 1 refers to the average of C_d under the condition of level 1, and so as the Mean 2 to Mean 5. Range refers to

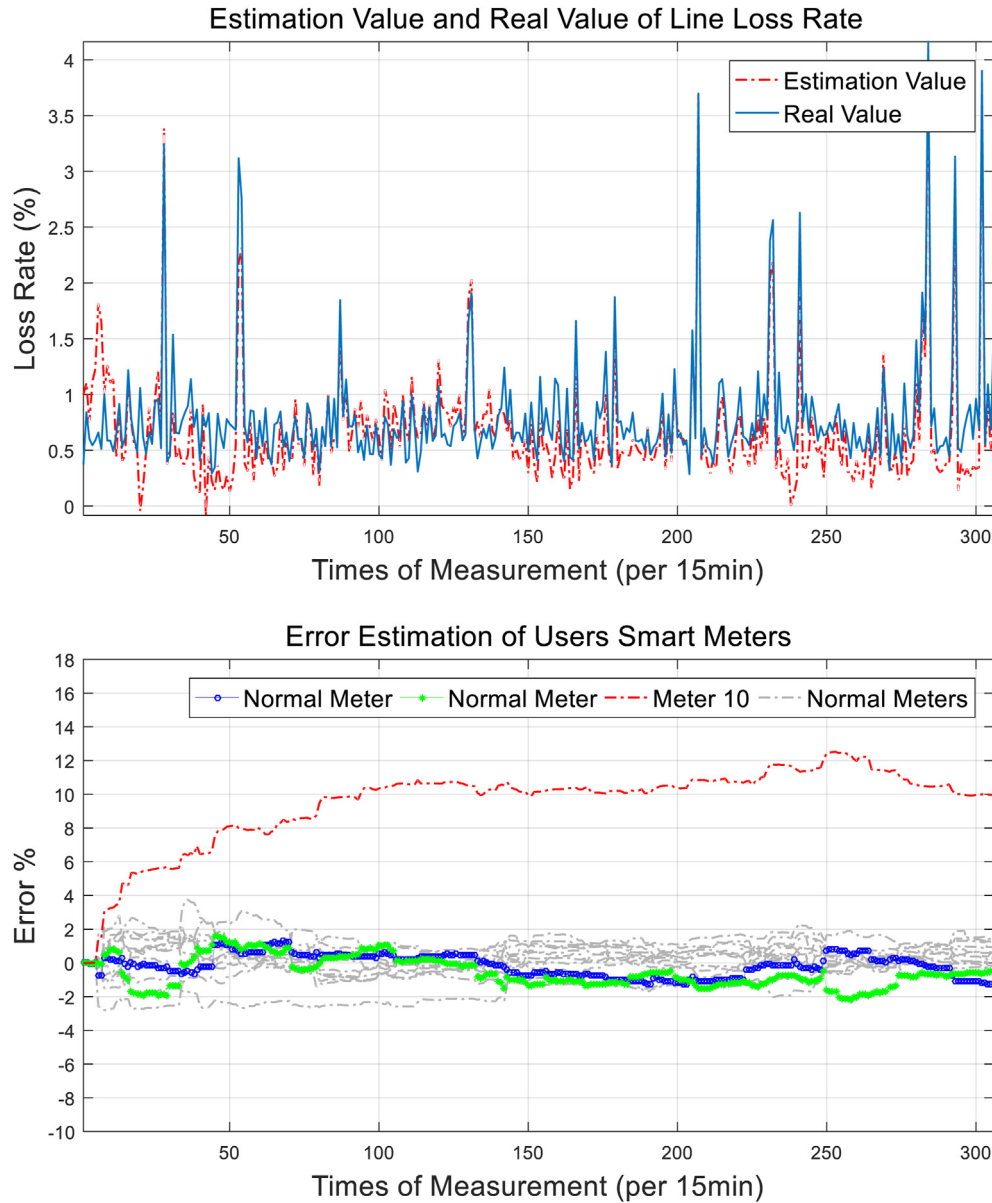


Fig. 10. The estimation results of proposed method (PL).

the difference between max and min of Mean. T_{Cd} refers to the sum of C_d , T_{mean} refers to the sum of the Mean. The relationship between T_{Cd} and T_{mean} is $T_{Cd} = 5 \times T_{mean}$. Among these factors, factors D, E and F are reserved to determine the sum of dispersion squares

Table 1

The index calculating results of the methods.

Model	Indices	RLS	DRLS
DREM-CR (Proposed model)	D_x (%)	1.3253	0.8335
	D_{loss} (%)	0.2539	0.1909
	C_d (%)	61.7356	91.8301
	D_x (%)	1.4165	0.9193
DREM-PL (Proposed model)	D_{loss} (%)	0.2886	0.2528
	C_d (%)	55.5556	81.3725
	D_x (%)	5.5231	4.3772
	D_{loss} (%)	1.9100	1.2117
KEM	C_d (%)	48.0392	50.9804
	D_x (%)	13.6029	4.9325
	C_d (%)	15.3795	32.0261
CNLREM	D_x (%)		
	C_d (%)		

P.S. Since the CNLREM does not estimate the network loss rate, the CNLREM does not participate in the discussion of the RMSE of the linear loss rate.

caused by experiment random errors. From the Range, it can be seen that factor A (i.e., the access to $\mathbf{M}(t)$) has the greatest influence on the success rate of estimation, followed by factor C (i.e., damping factor μ_0), and finally factor B (i.e., cluster number s). It can be found that the Range of factors D, E and F is not low, which indicates that the experiment random errors are large. It is supposed to be caused by the random selection of clustering center of k-means clustering program of MATLAB.

The effect curve of the orthogonal experiment is shown in Fig. 12. It can be seen that for factor A (i.e., the access to $\mathbf{M}(t)$), the success rate of estimation is the highest in the case of level 1 and level 5 (i.e., clustering extended measurement item $\mathbf{R}_{ex}(t)$), so this level is recommended. For factor B (i.e., cluster number s), the success rate of estimation is the highest in the case of level 4 (i.e., $s = 20$), and this level is recommended. For factor C (i.e., damping factor μ_0), the success rate of estimated at level 2 is slightly higher than that at level 3. It has been explained in section 3.3 that the greater the damping factor is, the more times recursion convergence is required. Therefore, it is recommended to use level 3 in the

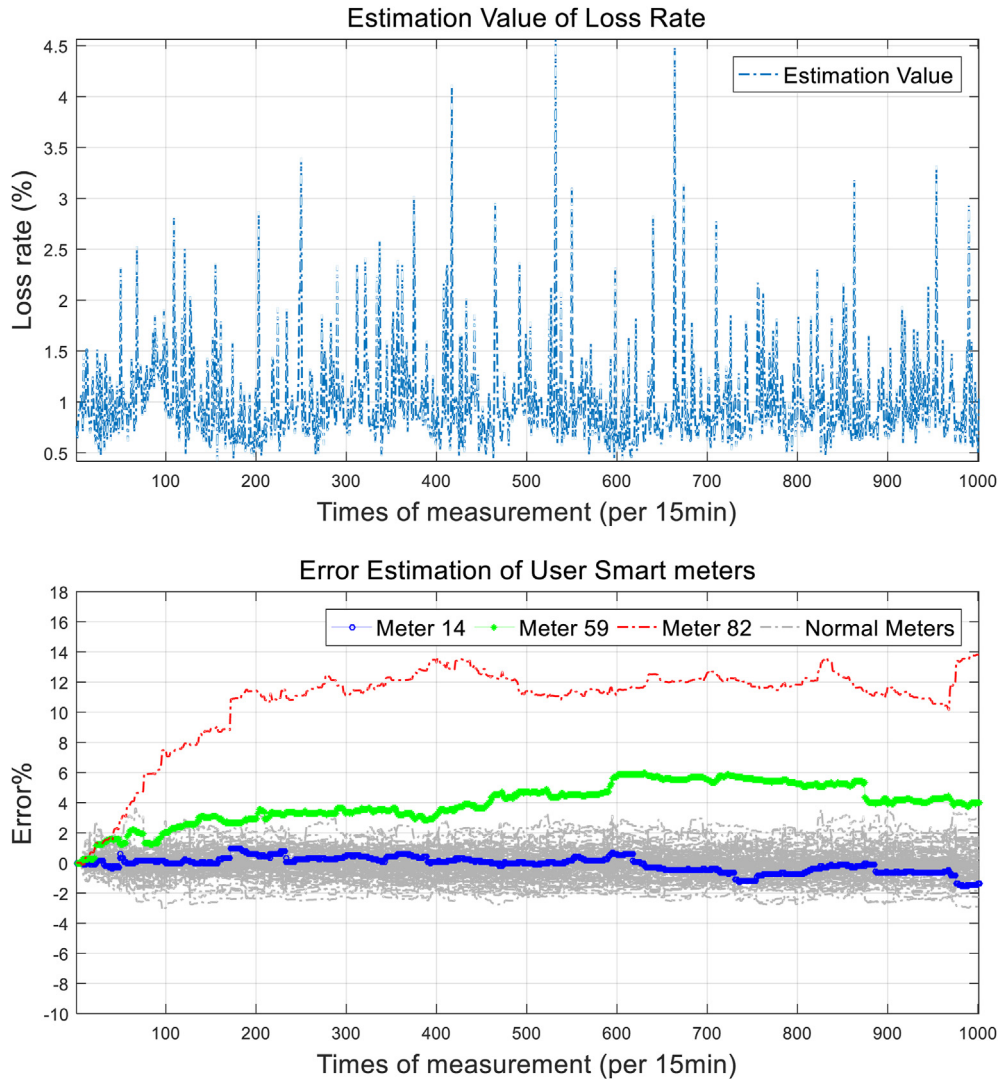


Fig. 11. The estimation results of actual distribution feeder unit.

Table 2

The index calculating results of different models in actual distribution feeder unit.

Model \ Indices	D_x (%)	C_d (%)
Proposed model	0.8161	72.6228
KEM	2.0713	32.1520
CNLREM	5.9501	10.654

Table 3

The rates of omission and undue-detection of different models.

Methods	Proposed model	KEM	CNLREM
Num. of omission	0	0	0
Num. of undue-detection	5	37	71
Rate of omission	0%	0%	0%
Rate of undue-detection	4.46%	33.03%	63.39%

case where the real-time requirement of the smart meter error estimation is high, otherwise level 2 should be used. To sum up, the parameter optimization selection presented in this paper is: (1) clustering the extended measurement item to $\mathbf{R}_{ex}(t)$ obtain the membership matrix; (2) cluster numbers = 20; (3) damping factor $\mu_0 = 0.01$.

Table 4

Orthogonal Experiment: correspondence between factors and levels.

Factor	A	B	C
Level	access to $\mathbf{M}(t)$	s	μ_0
1	Clustering $\mathbf{R}_{ex}(t)$	1	10
2	Clustering $\mathbf{Y}(t)$	5	1
3	Clustering $\mathbf{i}(t)$	10	10^{-2}
4	Clustering $\mathbf{Y}_d(t)$	20	10^{-4}
5	Clustering $\mathbf{R}_{ex}(t)$	40	10^{-6}

4.3.4. The influence of load condition

As the network loss plays an important role on the remote estimation of smart meter errors, the proposed method needs to consider the frequency and degree of network loss changes, which depends on both parameters of network and load condition. Since the network parameters of actual distribution feeder units are hardly accessible, the influence of network loss is indirectly investigated by studying the performance in the case of light load and heavy load. The average value of the network loss rate from the estimation in the former case was 1.1%. While the network loss rate in the case would be 1.9% by selecting the measurement data from

Table 5
Orthogonal experimental results.

Factor	A	B	C	D	E	F	$C_d(\%)$
Test 1	1	1	1	1	1	1	69.3674
Test 2	1	2	2	2	2	2	80.7991
Test 3	1	3	3	3	3	3	88.7902
Test 4	1	4	4	4	4	4	88.6792
Test 5	1	5	5	5	5	5	88.0133
Test 6	2	1	2	3	4	5	25.0832
Test 7	2	2	3	4	5	1	32.626
Test 8	2	3	4	5	1	2	14.0954
Test 9	2	4	5	1	2	3	10.8768
Test 10	2	5	1	2	3	4	0
Test 11	3	1	3	5	2	4	10.9878
Test 12	3	2	4	1	3	5	10.8768
Test 13	3	3	5	2	4	1	7.8801
Test 14	3	4	1	3	5	2	0
Test 15	3	5	2	4	1	3	26.3041
Test 16	4	1	4	2	5	3	0.222
Test 17	4	2	5	3	1	4	0.333
Test 18	4	3	1	4	2	5	1.2625
Test 19	4	4	2	5	3	1	19.167
Test 20	4	5	3	1	4	3	0.9469
Test 21	5	1	5	4	3	2	2.2198
Test 22	5	2	1	5	4	3	34.2952
Test 23	5	3	2	1	5	4	74.6948
Test 24	5	4	3	2	1	5	91.6759
Test 25	5	5	4	3	2	1	87.0144
Mean 1	83.130	21.576	20.985	33.353	40.355	43.211	$T_{Cd} = 866$
Mean 2	16.536	31.786	45.21	36.115	38.188	24.279	$T_{mean} = 173$
Mean 3	11.21	37.345	45.005	40.244	24.211	26.906	
Mean 4	4.386	42.08	40.178	30.218	31.377	34.939	
Mean 5	57.98	40.456	21.865	33.312	39.111	43.382	
Range	78.744	20.504	24.225	10.026	16.144	19.103	

months that were supposed to be heavy loading. The DREM-CR was applied with parameter setting as recommended. The results are shown in Fig. 13 and Table 6.

According to the results, the larger the network loss rate, the worse the estimation results would be. When the average network loss rate changed to twice the original one, $D_x(\%)$ increased by 40%, $C_d(\%)$ decreased by 20%, and the rate of undue-detection become twice. However, the results were still acceptable, which verified the practicability of proposed method.

5. Conclusion

To solve the problem caused by on-site calibration and periodic replacement of smart meters, the remote estimation method of smart meters errors based on DREM and DRLS is proposed, which could estimate the network loss rate and smart meter errors in real-time under the present measurement system. The followings conclusions are obtained:

- (1) Two DREM building options are proposed. Option CR can realize higher accuracy of the estimation but requires current measurement of the feeder meter. Though the estimation performance of Option PL is slightly lower than Option CR, but Option PL can be directly applied to present network. By applying k-means clustering, DREM can reduce the unknowns and transfer the unsolvable actual model into a solvable model.
- (2) By adding damping item to RLS, DRLS algorithm reduces the variation range of estimation values, which makes the estimation resistant to approximation errors and the variation of network loss, so the stability, robustness and accuracy are greatly improved. Parameter optimization is important for the full use of the proposed method, which includes the access of membership matrix, the clustering number and the damping factor, and the effect of parameter combination could obtain with the orthogonal experiment.
- (3) The proposed method was applied in both laboratory and actual distribution feeder units, which has acceptable and practical estimation results and superior to other methods. By applying the proposed method, the variation of smart meters state can be monitored timely, and online detection of power theft and leakage can be realized. The out of tolerance user meters can be located in advance, thus the workload of inspectors could be greatly reduced.
- (4) The application of the proposed method also has certain limitations, such as, linearization of nonlinear models is used in the estimation methods, which makes the problem easier to analyze and solve, the approximation may lead to deviations to a certain extent in the estimation results of smart meters. The k-means program in MATLAB is adopted for the clustering. Since the clustering center of this method is

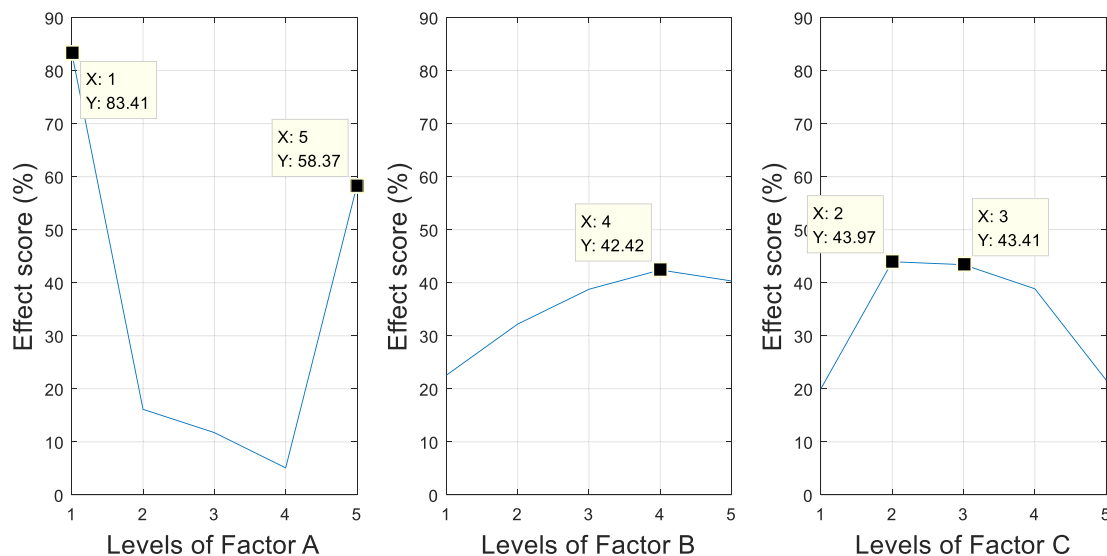


Fig. 12. Parametric effect curves.

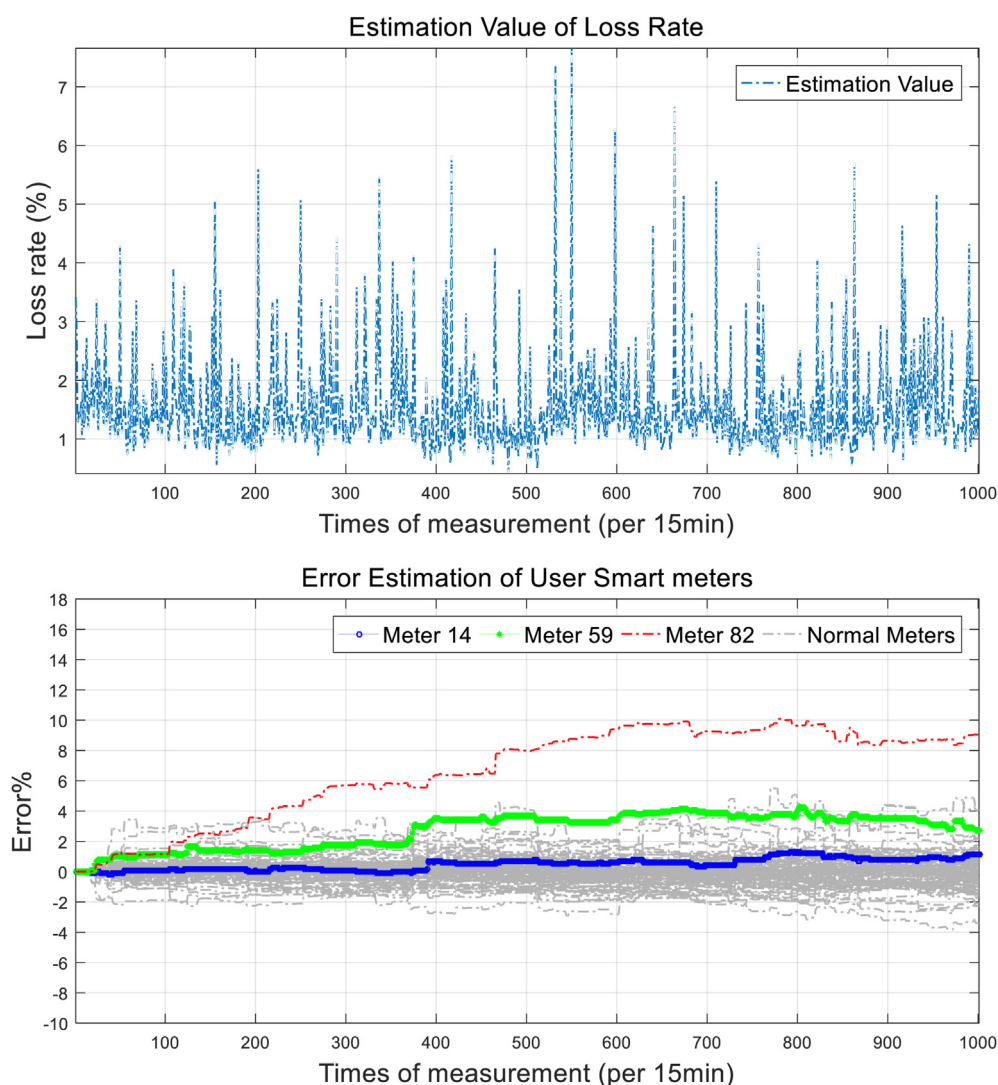


Fig. 13. The estimation results in the case of heavy load.

Table 6

The indices calculation in the case of heavy load.

Loss rate (%)	D_x (%)	C_d (%)	Omission	Undue-detection
1.1	0.8161	72.6228	0%	4.46%
1.9	1.1556	54.7032	0%	8.93%

determined randomly, it has a certain influence on the estimation results and indices calculation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2020.117774>.

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